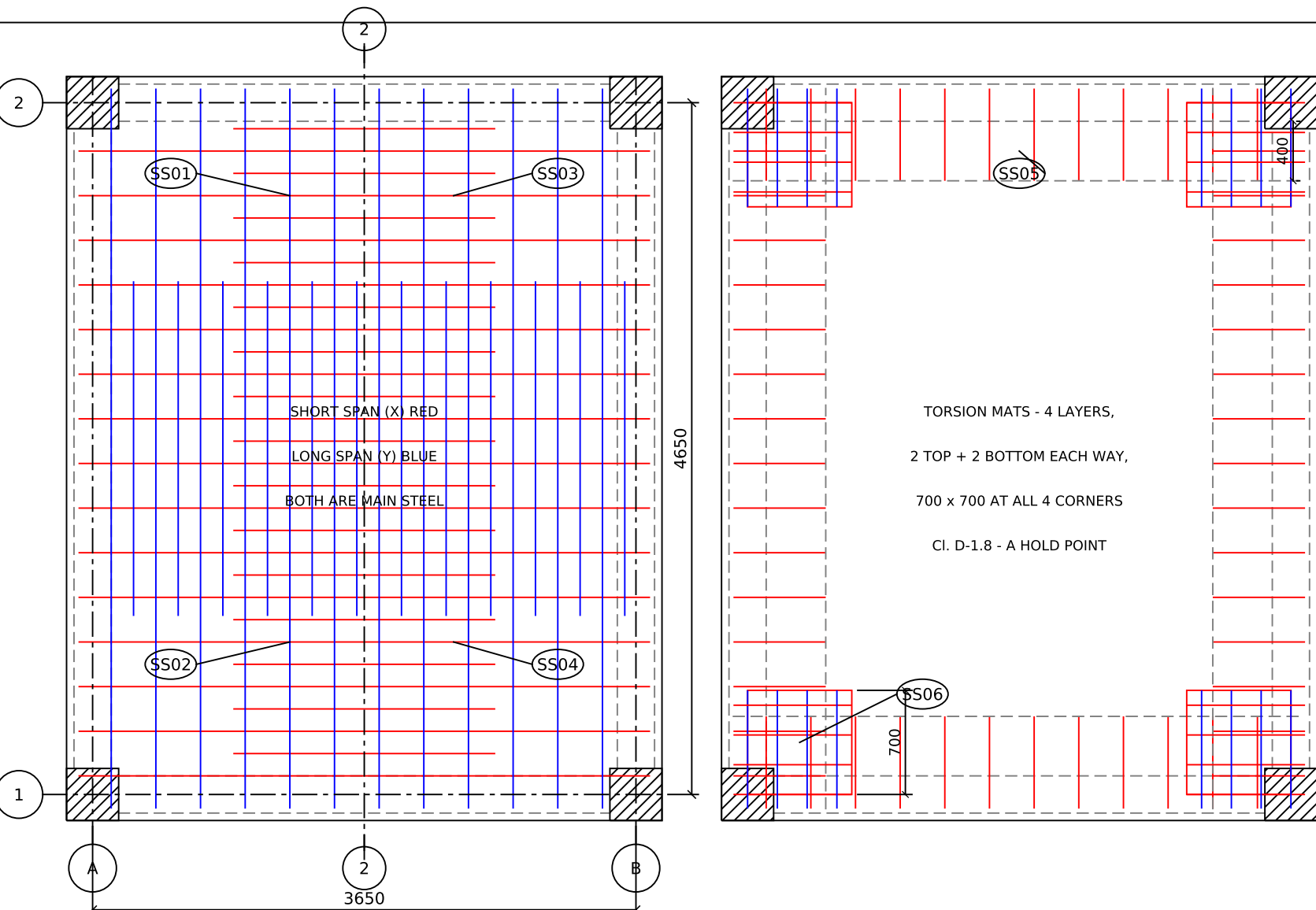
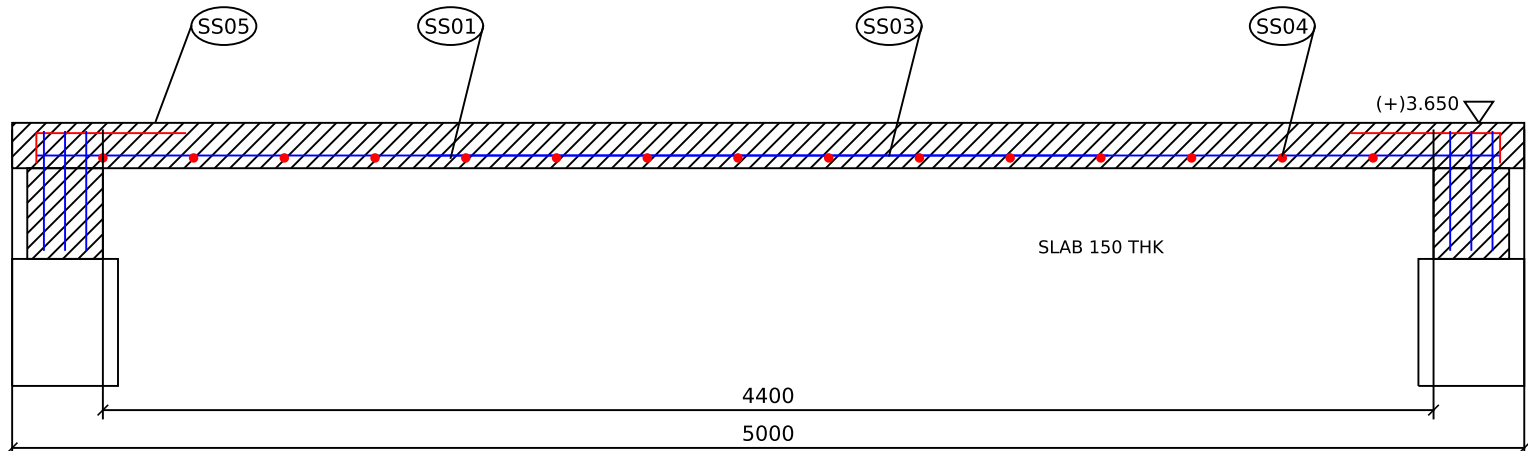


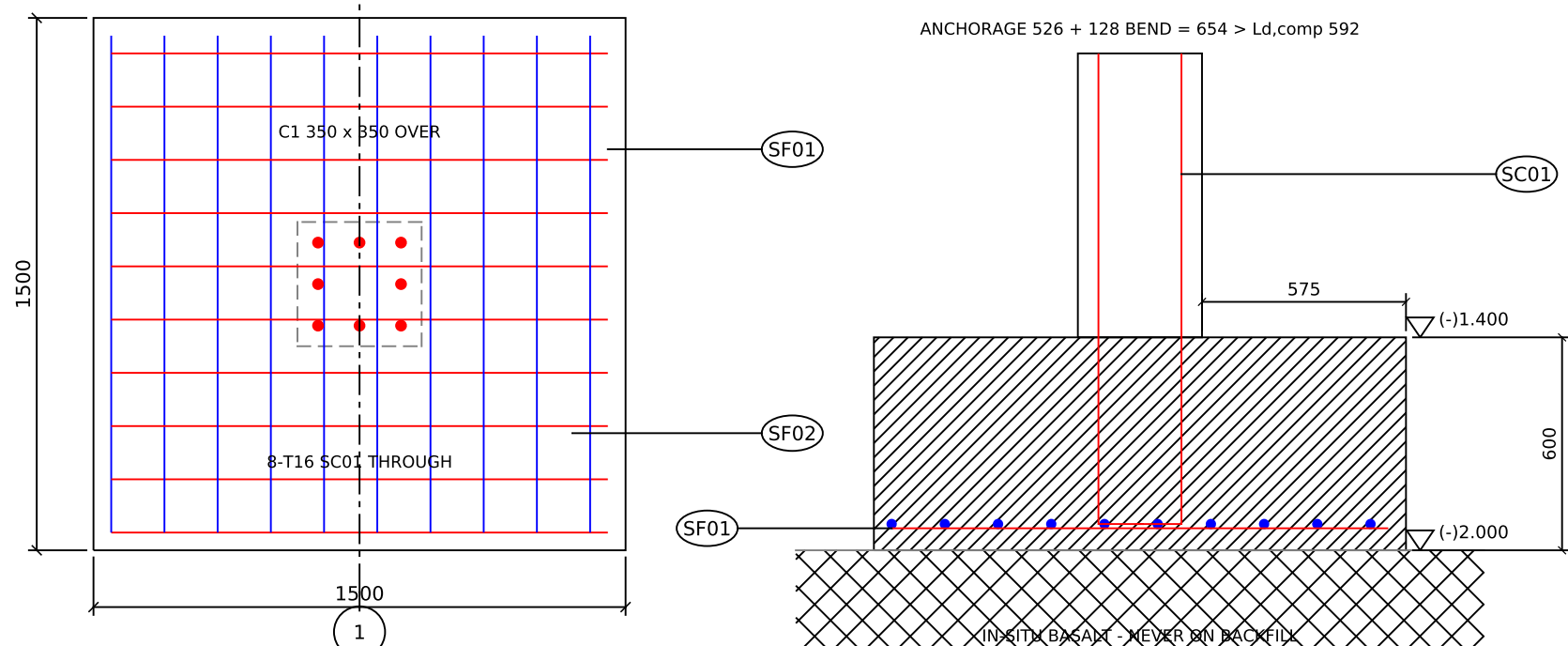


1 SLAB S1 - BOTTOM REINFORCEMENT PLAN 1 : 40 T8 @ 150 BOTH WAYS = 335 mm²/m, COVER 30

2 SLAB S1 - TOP REINFORCEMENT PLAN 1 : 40 400 EDGE BANDS (CI. D-1.6) + 700 x 700 CORNER MATS (CI. D-1.8)



3 SLAB S1 - LONGITUDINAL SECTION 2-2 1 : 25 TYPICAL AT FIRST FLOOR AND ROOF. B1 250 x 450 AT EACH END - SEE SHEET 08



4 ISOLATED FOOTING F1 - REINFORCEMENT PLAN 1 : 20 T12 @ 150 BOTH WAYS BOTTOM (754 mm²/m), COVER 50, 4 No.

5 ISOLATED FOOTING F1 - SECTION 1-1 1 : 20 600 DEPTH SET BY STARTER ANCHORAGE, NOT BY BENDING OR SHEAR

FOOTING SCHEDULE						
MARK	No.	SIZE	FOUNDING	TOP	COVER	d
F1	4	1500 x 1500 x 600	(-)2.000	(-)1.400	50	542
FOUNDED ON IN-SITU BASALT. PRESUMPTIVE SBC 3240 kPa, IS 1904 TABLE 1 [ASSUMED]						
SERVICE q _{max} 157.6 kPa = 4.9 % OF SBC e = 0.123 < L/6 = 0.250, NO TENSION						
THE 600 DEPTH IS SET BY STARTER ANCHORAGE, NOT BY BENDING OR SHEAR						

SLAB S1 - ELEMENT SCHEDULE					
ELEMENT	THK	COVER	d x / d y	REINFORCEMENT	BASIS
S1 BOTTOM, BOTH WAYS	150	30	116 / 108	T8 @ 150 (335 mm ² /m)	IS 456 Cl. 26.5.2.1 min 180
S1 TOP, DISCONTINUOUS EDGES	150	30	116	T8 @ 300, 400 INTO THE SPAN	Cl. D-1.6, all four edges
S1 CORNER TORSION MATS	150	30	-	T8 @ 200, 4 LAYERS, 700 x 700	Cl. D-1.8, all four corners
S1 OCCURS TWICE - FIRST FLOOR AND ROOF (MASTER A.4.8). THE FIRST FLOOR GOVERNS.					

BAR MARK SCHEDULE - FOOTING F1 AND SLAB S1					
MARK	BAR	SPACING	LOCATION	CUT mm	No.
SF01	T12	150	BOTTOM, BOTH WAYS - LAYER 1 (X)	1400	40
SF02	T12	150	BOTTOM, BOTH WAYS - LAYER 2 (Y)	1400	40
SS01	T8	300	BOTTOM, SHORT SPAN (X), FULL LENGTH (D4)	3840	30
SS02	T8	300	BOTTOM, SHORT SPAN (X), CURTAILED AT 0.25 L	1760	30
SS03	T8	300	BOTTOM, LONG SPAN (Y), FULL LENGTH (D4)	4840	24
SS04	T8	300	BOTTOM, LONG SPAN (Y), CURTAILED AT 0.25 L	2250	24
SS05	T8	300	TOP, 400 EDGE BAND, ALL FOUR EDGES	720	108
SS06	T8	200	CORNER TORSION MATS, 4 LAYERS, 700 x 700	795	128

IS 456 ANNEX D - TWO-WAY SLAB DETAILING		
CLAUSE	REQUIREMENT	AS DETAILED
D-1.1 / Table 26	CASE 9, FOUR EDGES DISCONTINUOUS BUT RESTRAINED	alpha x 0.0777, alpha y 0.056
C1. 24.4	ly/lx = 4508/3516 = 1.282 < 2 -> TWO-WAY	CONFIRMED
D-1.2 / D-1.3	MIDDLE STRIP 3/4, EDGE STRIPS 1/8 EACH SIDE	TABLE 26 MOMENTS APPLY TO THE MIDDLE STRIPS ONLY
D-1.4	50 % TO WITHIN 0.1 L OF A DISCONTINUOUS EDGE	SS01 / SS03 FULL LENGTH, SS02 / SS04 CUT AT 0.25 L
D-1.6	TOP STEEL = 50 % OF MIDSPAN, 0.1 L = 400 INTO THE SPAN	SS05 T8 @ 300, ALL FOUR EDGES
D-1.7	EDGE STRIPS - MINIMUM REINFORCEMENT ONLY	T8 @ 150 THROUGHOUT COVERS IT
D-1.8	CORNER TORSION = 3/4 OF MIDSPAN, 4 LAYERS OVER lx/5 = 700	SS06 T8 @ 200 - A HOLD POINT
Cl. 26.5.2.2	MAX BAR DIA D/8 = 18.75	T8 OK
Cl. 26.3.3(b)	MAX SPACING 3d OR 300	150 BOTTOM, 300 TOP OK

DECLARED DETAILING DECISIONS AND OPEN ITEMS

D4 SLAB BOTTOM STEEL. Master B.8.3 reads IS 456 Cl. D-1.4 as '50 % of midspan bottom steel extends to within 0.1 L of a discontinuous edge; remainder curtailed at 0.25 L'. The 50 % that continues (SS01 / SS03) is detailed FULL LENGTH into the beams - a bar that runs through certainly extends to within 0.1 L - and the remainder (SS02 / SS04) stops 0.25 L short of each support together they give T8 @ 150 both ways.

D3 THE COLUMN VERTICALS SC01 ARE CONTINUOUS FROM THE FOOTING TO THE ROOF. ONE 8 724 mm STOCK BAR, NO SPLICE. They are detailed on SHEET 08; only their anchorage inside F1 is drawn here. Master B.8.7: 526 straight + a 128 (8 phi) bend = 654 > Ld,comp 592 for T16 in M30.

SR2-V4 The footing TOP level foundation level (-)1.400 is the founding level (-)2.000 plus the 600 thickness. The master states the founding level only. [RECONSTRUCTED] Confirm against the setting-out before F1 is cast.

SR2-V5 NO blinding or levelling course under F1 is stated anywhere in the master, and none is drawn. [NOT AVAILABLE]

SR2-V6 IS 13920 Cl. 8.1 also asks for confining reinforcement to continue INTO the footing. Master F.4 states confining steel '500 from every joint face, top and bottom of every column, and through the joint' and stops there. The rule is applied as written and NO footing embedment is drawn. [UNRESOLVED - refer to the design authority before the footings are cast.]

DESIGN BASIS - THIS SHEET

ISOLATED FOOTING F1 1500 x 1500 x 600 ON IN-SITU BASALT AT (-)2.000 d = 600 - 50 - 8 = 542

*** FOUNDED ON IN-SITU ROCK, NEVER ON BACKFILL. This is the whole reason the sentry post is sited at least 10 m clear of the shelter excavation. ***

SERVICE P = 167.5 (column) + 36.1 (EQ axial) + 33.75 (footing self) = 237.4 kN ; M at the base 29.3 kNm

e = 29.3/237.4 = 0.123 m < L/6 = 0.250 -> NO TENSION UNDER THE BASE

q_{max} = P/A (1 + 6e/L) = 105.5 x 1.494 = 157.6 kPa vs SBC 3240 kPa (IS 1904 Table 1) = 4.9 %

ULS 1.5(DL + EL): Pu 276.8 kN ; Mu 43.9 kNm ; eu 0.128 ; q_{u,max} = 123.0 x 1.544 = 190.0 kPa

FLEXURE cantilever from the column face = (1500 - 350)/2 = 575 ; Mu = 190.0 x 0.575p/2 = 31.4 kNm/m

Ast,req 136 ; Cl. 26.5.2.1 MINIMUM 0.12 % x 600 = 720 mm²/m *** GOVERNS ***

PROVIDED T12 @ 150 BOTH WAYS (754), x/d 0.056

ONE-WAY SHEAR Cl. 34.2.4.1, critical at d from the face = 33 mm from the edge ; Vu 6.3 kN/m ; tau_v 0.012

PUNCHING Cl. 31.6.1, perimeter at d/2 : b0 = 4 x (350 + 542) = 3568 ; Vu = 276.8 - 190.0 x 0.892p = 125.6

tau_v 0.065 ; ks 1.0 ; tau_c = 0.25 sqrt(30) = 1.369 -> NOT CRITICAL

ANCHORAGE column starters T16, Ld,comp = 37 x 16 = 592. Available 526 straight + 128 (8 phi) bend = 654

*** THE 600 DEPTH IS GOVERNED BY STARTER ANCHORAGE, NOT BY BENDING OR SHEAR - so nobody optimises it. ***

SLAB S1 150 THK TWO-WAY, 3650 x 4650 c/c, BOTH FLOORS dx = 116 ; dy = 108

EFFECTIVE SPANS Cl. 22.2(a), lesser of clear + d or c/c : lx = min(3400+116, 3650) = 3516 ; ly = min(4400+108, 4650) = 4508. ly/lx = 1.282 < 2 -> TWO-WAY, Cl. 24.4 and Annex D

MOMENT COEFFICIENTS Table 26 (D-1.1) CASE 9, four edges discontinuous but RESTRAINED, with corner torsion steel to Cl. D-1.8 -> Table 26, NOT Table 27. alpha x 0.0777 (at 1.282), alpha y 0.056

wu = 1.5(4.750 + 3.000) = 11.625 kPa THE FIRST FLOOR GOVERNS (the roof is 10.125)

Mx = 0.0777 x 11.625 x 3.516p = 11.17 kNm/m ; My = 0.0560 x 11.625 x 3.516p = 8.05 kNm/m

Ast,req x 229 y 176 ; Cl. 26.5.2.1 minimum 0.12 % x 150 = 180 mm²/m

PROVIDED T8 @ 150 BOTH WAYS BOTTOM = 335 mm²/m -> Mu 16.09 / 14.92 kNm/m, UTILISATION 69 % / 54 %

DEFLECTION lx/dx = 30.3 ; basic taken as SIMPLY SUPPORTED = 20, the conservative assumption

fs 198 ; pt 0.289 % ; MF about 1.68 -> permissible 33.6 > 30.3 OK

SHEAR Vu about wu lx/3 = 13.6 kN/m ; tau_v 0.117 ; tau_c 0.390 x k 1.30 = 0.507 OK

*** THE CORNER TORSION MATS ARE A HOLD POINT FOR REINFORCEMENT INSPECTION. Table 26 is valid ONLY because they are provided. Omit them and the slab must be reassessed on Table 27. ***

LOADS (A.7.7) slab dead 4.750 (floor) / 5.250 (roof) kPa ; imposed 3.000 (IS 875 Pt 2, OP) / 1.500 kPa

Load path to the beams, Cl. 24.5, 45 deg yield lines; peak intensity = w x 3.650/2 = w x 1.825 m

SHORT beams B1 -> TRIANGLE ; LONG beams B2 -> TRAPEZOID; w,eq = w,peak x (1 - 1/3 rp), r 1.274

Infill on the first-floor beams 13.000 kN/m. SP-B1: brick at about 20 kN/m³ over 0.190 x 2.600 gives about 9.9 kN/m, i.e. LIGHTER, so 13.000 and Vb = 73.18 kN stay conservative. THAT IS A DIRECTION, NOT A VERIFICATION - the re-run is open as WM-V6.

OPEN ITEMS SR2-V4 footing top level RECONSTRUCTED. SR2-V5 no blinding stated. SR2-V6 confining steel into the footing not stated. A 7.7 the roof projection term 3.037 kN/m is [NOT AVAILABLE]. U4 the sentry post SITE POSITION is [ASSUMED] (RC4-F3). Neither sheet depends on it.

UNDERGROUND CBRN-HARDENED
BLAST-RESISTANT PROTECTIVE
STRUCTURE + SENTRY POST
PUNE, MAHARASHTRA
STRUCTURAL - PHASE 2 REV A + M1

NOTES

- ALL DIMENSIONS ARE IN MM. ALL LEVELS ARE IN METRES RELATIVE TO FINISHED SITE GRADE 0.000, NEGATIVE DOWNWARDS.
- WRITTEN DIM TO BE FOLLOWED ONLY. DRG NOT TO BE SCALED.
- THIS DRG IS TO BE READ IN CONJUNCTION WITH ALL RELEVANT ARCH AND STR DRGS. SENTRY POST CONCRETE M30.
- REINFORCEMENT Fe500D TO IS 1786:2008. COVER 40 TO COLUMNS, 30 TO BEAMS AND SLABS, 50 TO FOOTINGS (IS 456 Cl. 26.4.2 / TABLE 16).
- FOOTINGS ARE FOUNDED ON IN-SITU BASALT AT (-)2.000, NEVER ON BACKFILL. THAT IS THE WHOLE REASON THE SENTRY POST IS SITED AT LEAST 10 m CLEAR OF THE SHELTER EXCAVATION.
- THE 600 FOOTING DEPTH IS GOVERNED BY THE ANCHORAGE OF THE COLUMN STARTERS, NOT BY BENDING OR SHEAR. IT MUST NOT BE 'OPTIMISED' TO 350.
- SLAB S1 IS A TWO-WAY SLAB ON IS 456 TABLE 26 (ANNEX D-1.1, CASE 9). TABLE 26 IS VALID ONLY BECAUSE THE CORNER TORSION MATS ARE PROVIDED. THE MATS ARE A HOLD POINT FOR REINFORCEMENT INSPECTION.
- THE SENTRY POST FRAME IS THE IS 13920:2016 ELEMENT OF THIS PROJECT. IT IS A SPECIAL MOMENT-RESISTING FRAME, R = 3.0, Ah = 0.100, DESIGNED ON THE STAAD BASE SHEAR Vb = 73.18 kN.
- CONTRACTOR TO CHECK AND VERIFY THE DRG BEFORE EXECUTION.

STRUCTURAL REINFORCEMENT DETAILING OF SENTRY POST - FOOTING & SLAB

CBRN HARDENED UG OPS ROOM

SHEET 09

Project number	01
Date	16 SEP 2026
Drawn by	SYN 01
Checked by	DR IR CHAUDHARI
STR009	
Scale	As indicated